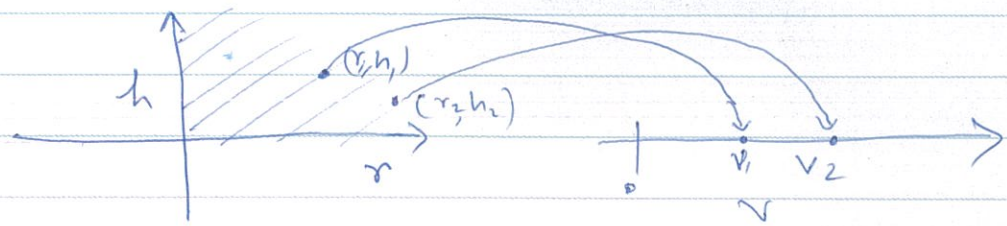
Partial Derivatives

Function of one variable : Area of a circle $A = \pi r^2$
where r is the radius

Function of two variables : volume of a cylinder $V = \pi r^2 h$
where h is the height of the cylinder

$\therefore$ Let $V = f(r, h) = \pi r^2 h$ where

$$f : (\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}) \to \mathbb{R}_{\geq 0}$$



~~Q:~~ Let $z = f(x, y) = \sqrt{y - x^2}$

Domain of f = $y \geq x^2$, Range $[0, \infty)$

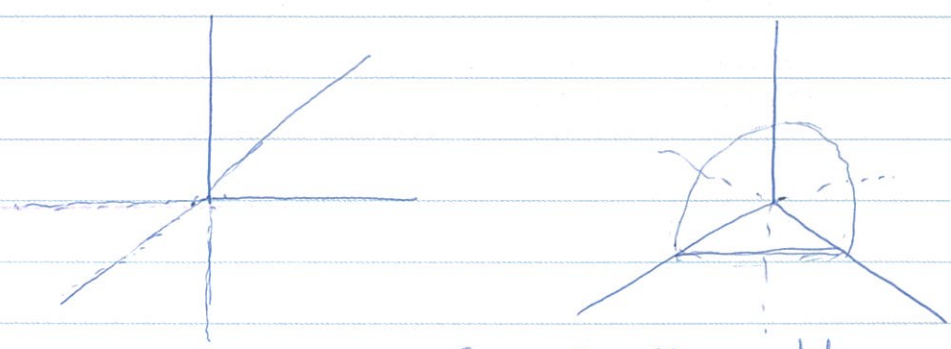
$w = f(x, y, z) = xy \log z$

Dom f = Half space $z > 0$, Range $(-\infty, \infty)$

Q: ? $z = \sqrt{1 - x^2 - y^2} = f(x, y)$

dom f = $0 \leq x^2 + y^2 \leq 1$, Rang $[0, 1]$

Now plot all the pts (x, y, z) on ~~x~~-y-z plane.



(upper) - Hemisphere.
Surface of the upper half of a ball.

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Limit

one var. fn

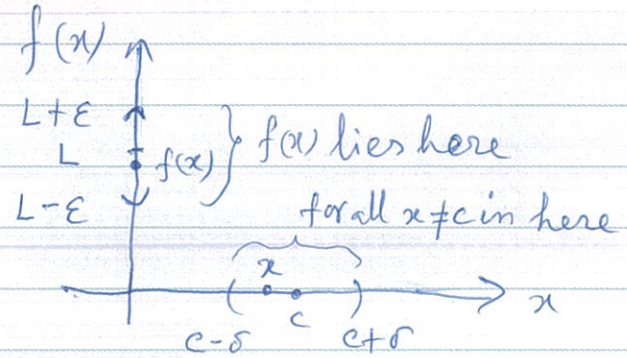
~~Def~~ Let $f(x)$ be defined on an open interval about c, except possibly at c itself. ~~we s~~

Def: We say that limit of $f(x)$ as x approaches c is the number L and write

$$\lim_{x \to c} f(x) = L \quad \text{if}$$

for every $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever } 0 < |x - c| < \delta.$$



For $f(x, y)$

Def: We say that a function $f(x, y)$ approaches the limit L as (x, y) approaches (x_0, y_0) and write

$$\lim_{(x,y) \to (x_0, y_0)} f(x, y) = L \quad \text{if,}$$

for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f

$$|f(x, y) - L| < \varepsilon \quad \text{whenever } 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$$

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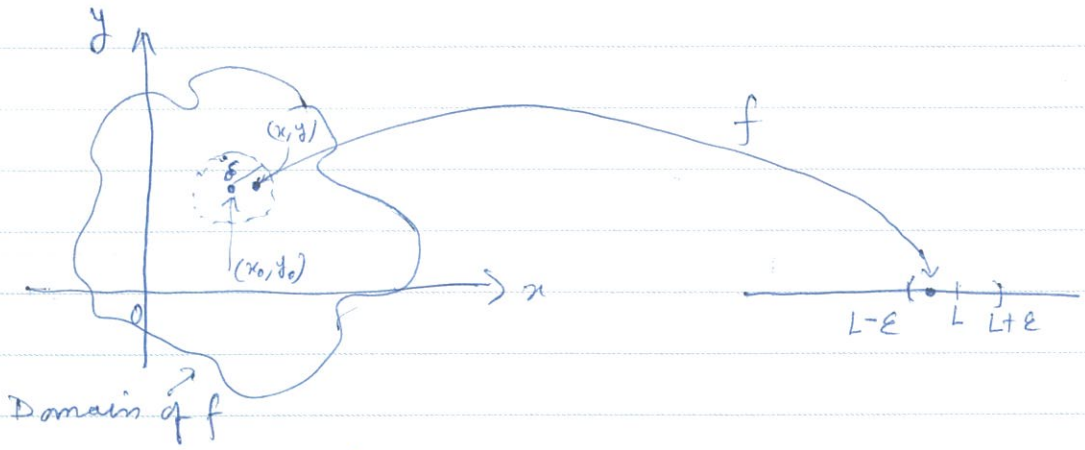

If $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = L$ and $\lim_{(x,y)\to(x_0,y_0)} g(x,y) = M$ then

$$\lim_{(x,y)\to(x_0,y_0)} \left(f(x,y) \pm g(x,y)\right) = L \pm M,$$

$$\lim_{(x,y)\to(x_0,y_0)} k f(x,y) = kL, \quad k \in \mathbb{R}$$

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) \cdot g(x,y) = L \cdot M$$

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) / g(x,y) = L/M, \quad \text{when } M \neq 0$$

$$\lim_{(x,y)\to(x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n},$$ where n is a +ive int. and if n is even, we assume $L > 0$.

<u>Ex-1</u> $$\lim_{(x,y)\to(0,1)} \frac{x - xy + 3}{x^2 y + 5xy - y^3} = \frac{0 - (0)(1) + 3}{(0)^2(1) + 5(0)(1) - (1)^3} = -3.$$

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Ex-2 $\lim_{(x,y)\to(0,0)} \dfrac{4xy^2}{x^2+y^2} = ?$

NB: The limit can't be found by direct substitution, which gives the indeterminate form $0/0$.

Solⁿ Observation: Along the line $x=0$, the fⁿ always has the value 0 when $y \neq 0$. Similarly, along the line $y=0$ when $x \neq 0$ the fⁿ has the value 0. So if the limit exists as (x,y) approaches $(0,0)$, the value of the limit must be 0. To see if this is true, we apply the def. of limit.

Let $\varepsilon > 0$ be given, but arbitary. Now we want to find a $\delta > 0$ such that

$$\left| \frac{4xy^2}{x^2+y^2} - 0 \right| < \varepsilon \quad \text{when} \quad 0 < \sqrt{x^2+y^2} < \delta$$

or.

$$\frac{4|x|y^2}{x^2+y^2} < \varepsilon \quad \text{when} \quad 0 < \sqrt{x^2+y^2} < \delta$$

Since $y^2 \leq x^2+y^2$ ~~the~~ we have

$$\left| \frac{4xy^2}{x^2+y^2} \right| = \frac{4|x|y^2}{x^2+y^2} \leq 4|x| = 4\sqrt{x^2} \leq 4\sqrt{x^2+y^2} \quad \left[\because \frac{y^2}{x^2+y^2} \leq 1\right]$$

So if we choose $\delta = \varepsilon/4$ and $0 < \sqrt{x^2+y^2} < \delta$ then

$$\left| \frac{4xy^2}{x^2+y^2} - 0 \right| \leq 4\sqrt{x^2+y^2} < 4\delta = 4(\varepsilon/4) = \varepsilon.$$

So from def. of lim we have

$$\lim_{(x,y)\to(0,0)} \frac{4xy^2}{x^2+y^2} = 0.$$

Two path test for non-existance of a limit: If a function $f(x,y)$ has different limits along two different paths in the domain of f as (x,y) approaches to (x_0,y_0), then $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ does not exist.

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Ex-3: Show that the function $f(x,y) = \dfrac{2x^2y}{x^4+y^2}$ has no limit as (x,y) approaches to $(0,0)$.

Sol^n: The limit can not be found by direct substitution, which gives the indeterminate form $0/0$.
We examine the value of f along parabolic curves that end at $(0,0)$. // Along the curve $y = kx^2$, $x \neq 0$, the function has the constant value

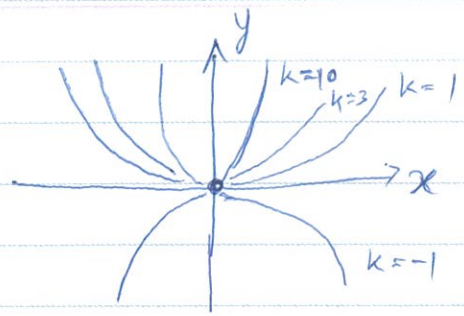
$$f(x,y)\Big|_{y=kx^2} = \frac{2x^2y}{x^4+y^2}\Big|_{y=kx^2} = \frac{2x^2(kx^2)}{x^4+(kx^2)^2} = \frac{2k}{1+k^2}$$

This limit varies with the path of approach. If ~~(0,0)~~ (x,y) approaches to $(0,0)$ along the parabola $y=x^2$, (when $k=1$) and the limit is 1. If (x,y) approaches to $(0,0)$ along the x-axis (i.e. $k=0$), the limit is 0. By two path test, f has no limit as (x,y) approaches $(0,0)$

Therefore, $$\lim_{\substack{(x,y)\to(0,0)\\ \text{along } y=kx^2}} f(x,y) = \lim_{(x,y)\to(0,0)}\left[f(x,y)\Big|_{y=kx^2}\right] = \frac{2k}{1+k^2}$$

Note: ① It can be shown that $f(x,y)$ has limit 0 along every st. line (i.e. path $y=kx$) [H.W.]

② Having the same limit along all st. lines approaching (x_0,y_0) does not imply that a limit exists at (x_0,y_0).



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